Monotonicity of Turán determinants for binary Krawtchouk polynomials

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Abstract. Let $M, Q \geq 0$ be integers, set $D = M + Q$, and define $p_s = [w^s]\,(1+w)^M(1-w)^Q$, with $p_s = 0$ outside $0 \leq s \leq D$. These are the binary Krawtchouk values $K_s(Q; D)$ in the generating-function normalization. We prove that the Turán determinants $\mathrm{Tur}(s) = p_s^2 - p_{s-1}p_{s+1}$ satisfy

$$\mathrm{Tur}(s) \geq \mathrm{Tur}(s+1) \qquad (\lceil D/2 \rceil \leq s \leq D).$$

Consequently, the determinants form a symmetric weakly unimodal sequence and satisfy $\mathrm{Tur}(s) \geq 1$ for every $0 \leq s \leq D$. By the dual Jacobi–Trudi identity, they are the rectangular Schur values $s_{(2^s)}(1^M, (-1)^Q)$; for $D \geq 1$, this gives positivity and unimodality for a family of general linear character values at involutions. The proof combines a quadratic-form identity with three comparison arguments for the three-term recurrence: a Riccati comparison near the top, a tangent comparison when the smaller exponent is even, and an interval propagation when it is odd. Four polynomial inequalities in the last argument are certified by coefficient positivity; no finite enumeration is needed. All results are formalized in Lean 4; the formalization, source programs and deterministic replay logs are available in the accompanying repository.

Keywords. Krawtchouk polynomials, Turán determinant, Schur function, character value, unimodality, three-term recurrence, exact computation

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1. Introduction and main theorem

1.1. The theorem

Fix integers $M, Q \geq 0$, set $D = M + Q$, and define the integer sequence

$$p_s \;=\; [w^s]\,(1+w)^M(1-w)^Q, \qquad s = 0, 1, \ldots, D, \tag{1.1}$$

with $p_s = 0$ outside $0 \leq s \leq D$. These are binary Krawtchouk values: with the standard generating function $\sum_k K_k(x; N)\, w^k = (1+w)^{N-x}(1-w)^x$ one has $p_s = K_s(Q; D)$, the degree-s binary (symmetric, $p = \frac{1}{2}$) Krawtchouk polynomial of length D evaluated at the integer point Q [KLS10, Section 9.11]. Define the *Turán determinant* and the *Turán step* of this sequence,

$$\begin{aligned} \mathrm{Tur}(s) &= p_s^2 - p_{s-1}p_{s+1}, \\ T_s &= \mathrm{Tur}(s) - \mathrm{Tur}(s+1) \;=\; p_s^2 + p_s p_{s+2} - p_{s+1}^2 - p_{s-1}p_{s+1}. \end{aligned} \tag{1.2}$$

Theorem 1.1 *(Main Theorem).* *For all integers $M, Q \geq 0$ and s with $\lceil D/2 \rceil \leq s \leq D$,*

$$T_s \;\geq\; 0.$$

Equivalently, $s \mapsto \mathrm{Tur}(s)$ is non-increasing on the top half. Since $p_{D-s} = (-1)^Q p_s$, the sequence $\mathrm{Tur}(s)$ is symmetric about $D/2$ and therefore weakly unimodal.

The following corollary is immediate by telescoping from the top end ($p_D = (-1)^Q$, $p_{D+1} = 0$, so $\mathrm{Tur}(D) = 1$).

Corollary 1.2. *For all $M, Q \geq 0$:*

1. *$\mathrm{Tur}(s) \geq 1$ for every $0 \leq s \leq D$, with sharp constant 1;*
2. *$\mathrm{Tur}(s) \leq \mathrm{Tur}(\lceil D/2 \rceil)$ for $0 \leq s \leq D$.*

For example, take $(M, Q) = (5, 2)$. Then

$$(p_s)_{s=0}^{7} = (1, 3, 1, -5, -5, 1, 3, 1), \qquad \big(\mathrm{Tur}(s)\big)_{s=0}^{7} = (1, 8, 16, 30, 30, 16, 8, 1).$$

Here the signs of p_s change while all Turán determinants are positive.

Classical Turán inequalities concern the sign of $P_n(x)^2 - P_{n-1}(x)P_{n+1}(x)$ for a fixed degree index n and an argument x in the orthogonality interval [Sze48]. Theorem 1.1 instead concerns monotonicity in the degree index (equivalently, the generating-function coefficient index) at a fixed integer argument. The integrality of the argument matters: if the coefficients are defined by the same formal series $[w^s](1+w)^{D-q}(1-w)^q$ for real $q \in [0, D]$, then $(D, q, s) = (4, \frac{8}{7}, 2)$ gives $T_2 = -1430/117649 < 0$. Related inequalities for binary Krawtchouk polynomials, their zeros, and their Christoffel–Darboux kernels were studied by Krasikov and Litsyn [KL96] and Krasikov [Kra01, Kra03, Kra05]; Section 8 compares them with Theorem 1.1.

1.2. Schur-character interpretation

Let

$$z \;=\; (\underbrace{1, \ldots, 1}_{M}, \underbrace{-1, \ldots, -1}_{Q}),$$

and write $e_r(z)$ and $s_\lambda(z)$ for the elementary and Schur symmetric functions, respectively. As usual, (2^s) denotes the rectangular partition with s rows of length 2.

Proposition 1.3 *(Schur-character interpretation).* *For every $0 \leq s \leq D$,*

$$\mathrm{Tur}(s) \;=\; s_{(2^s)}(z). \tag{1.3}$$

When $D \geq 1$, this is the irreducible polynomial-character value of $\mathrm{GL}_D(\mathbb{C})$

$$\chi_{(2^s)}(\mathrm{diag}(I_M, -I_Q))\,.$$

The rectangle-complement identity gives $s_{(2^s)}(z) = s_{(2^{D-s})}(z)$ independently of the Main Theorem.

Proof. The generating function in (1.1) is $\prod_{i=1}^{D}(1+z_i w)$, so $p_r = e_r(z)$. Since the conjugate of (2^s) is (s,s), the dual Jacobi–Trudi identity [Mac95, Chapter I, (3.5), p. 41] gives

$$s_{(2^s)}(z) = \det \begin{pmatrix} e_s(z) & e_{s+1}(z) \\ e_{s-1}(z) & e_s(z) \end{pmatrix} = p_s^2 - p_{s-1}p_{s+1}.$$

Schur polynomials are the irreducible polynomial characters of the general linear group; see [Mac95, Chapter I, Appendix A, (8.2) and (8.4), pp. 162–163]. Finally, for a partition λ inside the $2 \times D$ rectangle, with rectangle complement λ^c,

$$s_\lambda(z_1^{-1}, \dots, z_D^{-1}) = (z_1 \cdots z_D)^{-2} s_{\lambda^c}(z_1, \dots, z_D).$$

Here $z_i^{-1} = z_i$, $(z_1 \cdots z_D)^2 = 1$, and the complement of (2^s) is (2^{D-s}), proving the symmetry. ∎

Thus Theorem 1.1 and Corollary 1.2 say that the rectangular character values in (1.3) are positive integers and form a symmetric weakly unimodal sequence as the rectangle height s varies. For $Q > 0$, the diagonal matrix has negative eigenvalues, so Schur positivity does not by itself determine the sign.

1.3. Normalization

The normalization in (1.1) is part of the statement. If $q_s = c_s p_s$, then $q_s^2 - q_{s-1}q_{s+1} = c_s^2 p_s^2 - c_{s-1}c_{s+1}p_{s-1}p_{s+1}$, so a Turán determinant is merely rescaled by a positive factor for every sequence p only when (c_s) is geometric. Degree-dependent orthonormal or endpoint normalizations of Krawtchouk polynomials therefore do not preserve Theorem 1.1. The coefficient normalization is integral at integer arguments, has the reflection $p_{D-s} = (-1)^Q p_s$, and fixes $\mathrm{Tur}(D) = 1$; the last two features turn top-half monotonicity into symmetric weak unimodality and the sharp bound of Corollary 1.2.

1.4. Outline of the proof

The top half $\{2s \geq D\}$ is divided into the bulk, flow, center, and gap regions (Figure 1). A positive-semidefinite quadratic form treats the bulk (Theorem 3.1) and a Riccati comparison treats the flow region (Theorem 3.3). Lemma 3.6 confines the remaining gap to small offsets $u = 2s - D$, namely $6(2Q+3)(u+1)^2 < 7(D+3)$, where Q is the smaller exponent. Gap cells with Q even are treated by a tangent comparison for the ratio of consecutive coefficients (Theorem 4.1). For Q odd, the offset $u = 1$ has a closed form (Proposition 5.1), and larger offsets are treated by propagating a lower bound for that ratio through the recurrence (Theorem 5.3). Section 6 assembles the cases.

We assume $x = M - Q \geq 0$ without loss of generality. Indeed, T_s is invariant under $M \leftrightarrow Q$: the substitution $w \mapsto -w$ sends p_s to $(-1)^s p_s$, and every product in T_s has even index sum.

The algebraic identities below are replayed in exact rational arithmetic by the programs of Section 7. Apart from identities, the only inequalities established by computer are the four transition obligations $O_1, \dots, O_4$ of Lemma 5.2; each becomes a polynomial with nonnegative

coefficients after an explicit substitution. Every numbered statement and displayed claim of the paper is formalized in Lean 4 (Section 7.2).

2. Notation and recurrence

Throughout:

$$\begin{aligned}
&M, Q \in \mathbb{Z}_{\geq 0}; \quad D = M + Q; \quad x = M - Q \geq 0;\\
&p_s = [w^s]\,(1+w)^M(1-w)^Q \text{ (integers; } p_s = 0 \text{ off } [0, D]);\\
&s \text{ top-half index, } D/2 \leq s \leq D; \quad u = 2s - D \geq 0 \text{ (top-half offset)};\\
&U = (u+1)^2; \quad E = (D+3)^2;\\
&\mathrm{Tur}(s) = p_s^2 - p_{s-1}p_{s+1}; \quad T_s = \mathrm{Tur}(s) - \mathrm{Tur}(s+1);\\
&k = D - s \text{ (distance from the top } = (D-u)/2).
\end{aligned}$$

Reflection: $p_{D-j} = (-1)^Q p_j$ for all j (substitute $w \to 1/w$). The top initial values are $p_D = (-1)^Q$ and $p_{D-1} = (-1)^Q x$. Logarithmic differentiation of the generating function gives the step-1 recurrence

$$(s+1)\,p_{s+1} \;=\; x\,p_s + (s-1-D)\,p_{s-1}, \tag{2.1}$$

valid for every integer s. A *cell* is an integer triple (M, Q, s) with $\lceil D/2 \rceil \leq s \leq D$; we also write cells in the coordinates (D, x, u), which satisfy $u \equiv x \equiv D \pmod 2$.

Lemma 2.1 *(no consecutive zeros).* *For $0 \leq j \leq D$, the two consecutive values p_j, p_{j+1} cannot both vanish.*

Proof. If $j = 0$, the assumption contradicts $p_0 = 1$. If $1 \leq j \leq D$ and $p_j = p_{j+1} = 0$, then (2.1) at index j gives $p_{j-1} = 0$ because $j - 1 - D \neq 0$. Iteration gives $p_0 = 0$, again a contradiction. ∎

In Sections 4 and 5 we relabel the exponents when necessary so that $0 \leq Q \leq M$; thus Q denotes the smaller exponent.

3. Quadratic form and regime decomposition

3.1. The bulk band

Theorem 3.1 *(quadratic-form identity and the bulk band).* *Eliminating p_{s+2} and p_{s-1} from T_s by the step-1 recurrence (2.1) gives the identity*

$$\begin{aligned}
T_s \;&=\; A\,p_s^2 + B\,p_s p_{s+1} + C\,p_{s+1}^2,\\
A = \frac{u+2}{s+2}, \qquad B &= \frac{-x(u+1)}{(D+1-s)(s+2)}, \qquad C = \frac{u}{D+1-s},
\end{aligned} \tag{3.1}$$

with negated discriminant

$$4AC - B^2 \;=\; \frac{u(u+2)\big((D+3)^2 - (u+1)^2\big) - x^2(u+1)^2}{\big((D+1-s)(s+2)\big)^2}. \tag{3.2}$$

Hence for $u \geq 1$ (note $u(u+2) = U-1$):

$$x^2 U \;\leq\; (U-1)(E-U) \quad\Longrightarrow\quad T_s \geq 0. \tag{3.3}$$

Proof. Substitute (2.1) for p_{s+2} and p_{s-1} in (1.2); simplification gives (3.1) and (3.2). If $u \geq 1$, then $A, C > 0$, and the quadratic form is positive semidefinite whenever $4AC - B^2 \geq 0$. ∎

Remark. Expressing T_s in a shifted recurrence basis multiplies the negated discriminant by a positive square; in the basis (p_{s-2}, p_{s-1}), for instance, by $\big((D+1-s)(D+2-s)/(s(s+1))\big)^2$. No such change of basis enlarges the bulk band, so the remaining regions need further information from the recurrence.

3.2. The center

Theorem 3.2 *(central identity, $u = 0$, D even).* *By reflection ($p_{D-j} = (-1)^Q p_j$): for Q odd, $p_{D/2} = 0$ and $T_{D/2} = 0$ identically. For Q even, the reflection constraint $p_{s-1} = p_{s+1}$ at $s = D/2$ collapses the recurrence to the rank-one relation $(D+2)\,p_{s+1} = x\,p_s$, and*

$$T_{D/2} \;=\; \frac{4\,p_{D/2}^2\,\big((D+2)^2 - x^2\big)}{(D+2)^2(D+4)} \;>\; 0. \tag{3.4}$$

Proof. If Q is odd, reflection gives $p_{D/2} = 0$ and $p_{D/2-1} = -p_{D/2+1}$, which makes the four terms in $T_{D/2}$ cancel. If Q is even, reflection gives $p_{s-1} = p_{s+1}$ at $s = D/2$. Substitution in (2.1) yields $(D+2)p_{s+1} = xp_s$; inserting this relation into (1.2) and using the recurrence once more for p_{s+2} gives (3.4). Here $(D+2)^2 - x^2 \geq 4D + 4 > 0$, and $p_{D/2} \neq 0$ because otherwise the rank-one relation would give two consecutive zeros, contrary to Lemma 2.1. ∎

In particular, $T_{D/2} = 0$ exactly when M and Q are both odd.

3.3. The flow region: Riccati comparison

Theorem 3.3 *(flow region).* *Let $k = D - s \geq 1$. Whenever its radicand is nonnegative, let*

$$w_s \;=\; \sqrt{x^2 - 4k(D+1-k)}$$

denote the nonnegative square root. On the flow region $w_s^2 \geq 0$, equivalently $(u+1)^2 \geq (D+1)^2 - x^2$ by the identity $(D+1)^2 - (u+1)^2 = 4(s+1)(D-s)$, one has $T_s \geq 0$.

Proof. The flow region with $k \geq 1$ is empty when $x = 0$, so assume $x > 0$. On the top half, $0 \leq k \leq D/2$, and $k \mapsto 4k(D+1-k)$ is increasing. Hence if the target index is in the flow region, so is every index between it and $D-1$; the following downward induction runs on this connected tail. The center $u = 0$ cannot lie in the flow region, because there $4k(D+1-k) = D(D+2) > D^2 \geq x^2$. Thus $u \geq 1$ and the quadratic-form coefficient C is positive throughout the tail.

Use the ratio variable $r_s = p_{s+1}/p_s$. The top initial values $p_D = (-1)^Q$, $p_{D-1} = (-1)^Q x$ give $r_{D-1} = 1/x$, and the recurrence gives the downward Möbius flow

$$r_{s-1} \;=\; \frac{D+1-s}{x-(s+1)\,r_s}.$$

Set $\phi_s = (x+w_s)/2$ and define the auxiliary downward Riccati flow $y_{s-1} = x - (s+1)(D-s)/y_s$, $y_{D-1} = x$. Then:

1. *(comparison)* The inequality $\phi_{s-1} \le x - (s+1)(D-s)/\phi_s$ reduces, after clearing denominators by $4(s+1)(D-s) = x^2 - w_s^2$, to $(w_s - w_{s-1})(x+w_s) \ge 0$. This holds because $w_s^2 - w_{s-1}^2 = 4u > 0$. Induction from $y_{D-1} = x \ge \phi_{D-1}$ gives $y_s \ge \phi_s > 0$ on the connected tail.

   Starting from $r_{D-1} = 1/x = (D-(D-1))/y_{D-1}$, a simultaneous induction in the two displayed recurrences gives $r_s = (D-s)/y_s = k/y_s$. Consequently $0 < r_s \le k/\phi_s$, every Möbius denominator is positive, and no p_s in the tail vanishes.

2. *(window value)* The quadratic-form value of Theorem 3.1 at $r = k/\phi_s$ is $I_1/\big((s+2)(D+1-s)\phi_s^2\big)$ with

$$\begin{aligned} I_1 \;&=\; (u+2)(k+1)\,\phi_s^2 - x(u+1)\,k\,\phi_s + u(D-k+2)\,k^2 \;=\; P_1 + Q_1 w_s, \\ Q_1 \;&=\; \tfrac{1}{2}\,x(D-k+2) \;\ge\; 0, \\ P_1 \;&=\; \tfrac{1}{2}\big[x^2(D+2-k) - 2k(D^2 - 2Dk + 3D + 2k^2 - 2k + 2)\big]. \end{aligned} \tag{3.5}$$

   The coefficient $D+2-k$ of x^2 is positive. Thus P_1 is increasing in x^2, and at the flow boundary $x^2 = 4k(D+1-k)$ it equals $k(D+2)(u+1) > 0$. Since also $Q_1, w_s \ge 0$, this proves $I_1 \ge 0$ throughout the flow region.

3. *(window vertex)* Since $C > 0$, the inequality $k/\phi_s \le -B/(2C)$ reduces to

$$I_2 \;=\; x(u+1)\,\phi_s - 2u(D-k+2)\,k \;=\; P_2 + Q_2 w_s \;\ge\; 0, \tag{3.6}$$

   where

$$Q_2 \;=\; \frac{x(u+1)}{2}, \qquad P_2 \;=\; \frac{x^2(u+1) - 4k(D^2 - 3Dk + 2D + 2k^2 - 4k)}{2}.$$

   Since $u+1 > 0$, P_2 is increasing in x^2; at the flow boundary it equals $2k(k+1) > 0$. Hence $I_2 \ge 0$.

4. Since $A, C \ge 0 \ge B$, the quadratic $A + Br + Cr^2$ is decreasing on $[0, -B/(2C)]$; by (1)–(3), $T_s/p_s^2 = A + Br_s + Cr_s^2 \ge$ (value at k/ϕ_s) ≥ 0. ∎

3.4. The gap

Put

$$h \;=\; E + 1 - x^2, \qquad q(t) \;=\; t^2 - ht + E, \qquad U_0 \;=\; (D+1)^2 - x^2.$$

Expanding gives $x^2 U - (U-1)(E-U) = q(U)$, so the bulk inequality (3.3) fails exactly when $q(U) > 0$, and a cell with $k \ge 1$ lies outside the flow region exactly when $U < U_0$.

Corollary 3.4 *(coverage).* *For $D = 0$ the only cell is $s = 0$ and $T_0 = 1$. For $D \geq 1$, every top-half cell is covered by the top cell $s = D$ ($T_D = 1$), the center, the bulk, or the flow region, except the cells of the* gap

$$\mathcal{G} = \{u \geq 1 : \ q(U) > 0, \ U < U_0\}. \tag{3.7}$$

Moreover $q(U_0) \leq 0$, and $x > 0$ at every gap cell.

Proof. The first assertion follows from the two equivalences above. Since $E - U_0 = 4D + 8 + x^2$,

$$-q(U_0) = (U_0 - 1)(4D + 8) - x^2 = (D^2 + 2D - x^2)(4D + 8) - x^2,$$

which decreases linearly with x^2 and equals $7D^2 + 16D \geq 0$ at $x^2 = D^2$; hence $q(U_0) \leq 0$ for $0 \leq x \leq D$. If $x = 0$, then $q(U) = (U - 1)(U - E) > 0$ forces $U > E > U_0$, so $x = 0$ admits no gap cell. ∎

Lemma 3.5 *(heredity of the gap).* *If (D, x, u) is a gap cell and $u' \equiv u \pmod 2$ satisfies $1 \leq u' \leq u$, then (D, x, u') is also a gap cell.*

Proof. The parabola q has its vertex at $h/2$. If $U \geq h/2$, then q would be nondecreasing on $[U, U_0]$, and $q(U) \leq q(U_0) \leq 0$ would contradict $q(U) > 0$; hence $U < h/2$. Since q is decreasing on $(-\infty, h/2]$, $U' = (u' + 1)^2 \leq U$ gives $q(U') \geq q(U) > 0$, and $U' < U_0$. The parity condition makes $s' = (D + u')/2$ an integer index. ∎

Lemma 3.6 *(gap strip).* *After relabeling so that $Q \leq M$, every gap cell satisfies*

$$6(2Q + 3)(u + 1)^2 \ < \ 7(D + 3). \tag{3.8}$$

Proof. Here $h = (2Q + 3)(2M + 3) + 1$ and $h - 3(D + 3) = 4MQ + 3M + 3Q + 1 > 0$, so $h^2 > 9E$. Put $B = 7E/(6h)$. Then

$$q(B) = \frac{E\,(49E - 6h^2)}{36h^2} < 0.$$

If $U \geq B$, then $B \leq U < U_0$ with $q(B) < 0$ and $q(U_0) \leq 0$, and convexity of q gives $q(U) \leq 0$, a contradiction. Hence $6hU < 7E$. Finally $h - (2Q + 3)(D + 3) = (2Q + 3)x + 1 > 0$, so $6(2Q + 3)(D + 3)U < 6hU < 7E$; divide by $D + 3$. ∎

For $Q \geq 1$, (3.8) gives $30(u + 1)^2 < 7(D + 3)$. The first gap cell with Q odd and $u \geq 2$ is $(D, Q, u) = (84, 1, 2)$.

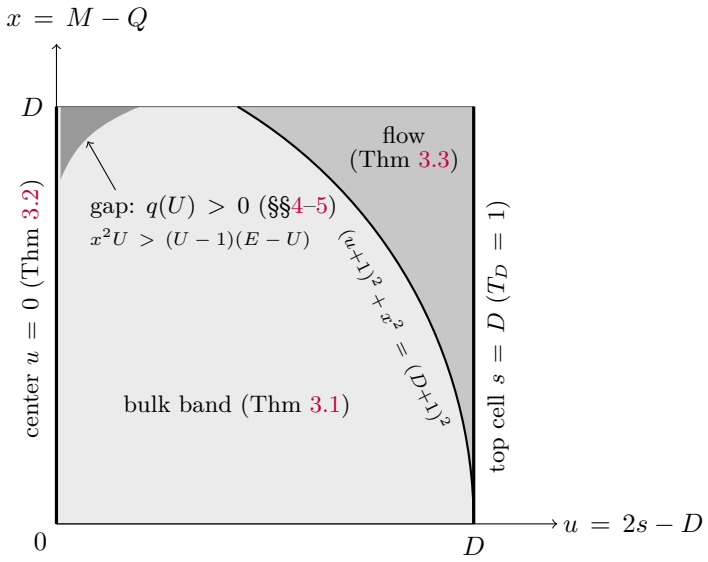


Figure 1. Coverage of the top half $\{2s \geq D\}$ in the (u, x)-plane for fixed D (schematic; lattice cells satisfy $u \equiv x \equiv D \pmod 2$). Cells in both the bulk and the flow region are drawn as flow, that is, beyond the displayed circle. The dark gap is the part of the non-flow region where the bulk quadratic form is indefinite; it is treated in Sections 4–5.

4. Gap cells with even Q

Theorem 4.1 *(gap cells with even Q). After relabeling so that $Q \leq M$, every gap cell with Q even has $T_s \geq 0$.*

Proof. For each top-half index j, define

$$d_j \;:=\; (D+2)\,p_{j+1} - x\,p_j \quad \text{(deviation from the central rank-one relation)},$$

$$\Lambda \;:=\; (D+2)^2 - x^2 \;=\; 4(Q+1)(M+1) \;>\; 0, \qquad \widehat{u}_j \;:=\; 2(j+1) - D.$$

By Lemma 3.5, every positive parity-compatible offset between the center and the target remains in the gap, and at the even-D center $(2j - D + 1)^2 = 1 < U < U_0$. Thus, for each index j used,

$$(2j - D)^2 \;<\; (2j - D + 1)^2 \;<\; (D+1)^2 - x^2.$$

We call the region $(2j - D)^2 < (D+1)^2 - x^2$ the *oscillatory zone*.

1. *(Turán-step identity)*

$$T_j \;=\; \frac{\widehat{u}_j\ \mathrm{Tur}(j)}{j+2} \;-\; \frac{p_{j+1}d_j}{(j+2)(D+1-j)}. \tag{4.1}$$

Since $\widehat{u}_j = 2(j+1) - D > 0$ at every top-half index used, $T_j \geq 0$ follows from $\mathrm{Tur}(j) \geq 0$ together with $p_{j+1}d_j \leq 0$.

2. *(positivity of the Turán determinant)*

$$(D+1-j)\,\mathrm{Tur}(j) \;=\; (D+1-j)\,p_j^2 - x\,p_j p_{j+1} + (j+1)\,p_{j+1}^2$$

is a quadratic form in (p_j, p_{j+1}) with discriminant $x^2 - \big[(D+2)^2 - (2j-D)^2\big]$ (using $4(j+1)(D+1-j) = (D+2)^2 - (2j-D)^2$), which is < 0 on the whole oscillatory zone. Therefore $\mathrm{Tur}(j) > 0$ there unless $(p_j, p_{j+1}) = (0,0)$, which is impossible by Lemma 2.1. The leading coefficient $D+1-j$ is positive because $j \le D$.

3. *((p,d) system)* The first-order recurrence for (p_j, d_j) is

$$p_{j+1} \;=\; \frac{x\,p_j + d_j}{D+2}, \qquad d_{j+1} \;=\; \frac{D-j}{(j+2)(D+2)}\,\big(x\,d_j - \Lambda\,p_j\big).$$

4. *(comparison map)* $\zeta_j := -d_j/p_j$ obeys

$$\zeta_{j+1} \;=\; \kappa_j\,\frac{\Lambda + x\,\zeta_j}{x - \zeta_j}, \qquad \kappa_j \;=\; \frac{D-j}{j+2}\ (\le 1 \text{ above the center});$$

the map is increasing in ζ on $[0,x)$ and in κ, and the $\kappa = 1$ map is exactly the tangent addition formula:

$$F\big(\sqrt{\Lambda}\,\tan a\big) \;=\; \sqrt{\Lambda}\,\tan(a+\theta^*), \qquad \tan\theta^* \;=\; \frac{\sqrt{\Lambda}}{x}.$$

Reflection supplies the initial values (Q even here, $p_{D-j} = p_j$). For D even, $d_c = 0$ at $c = D/2$ (the rank-one relation of Theorem 3.2), so $\zeta_c = 0$; here $p_c \neq 0$ by Theorem 3.2. For D odd, $p_{c+1} = p_c \neq 0$ at $c = (D-1)/2$ gives, in one step,

$$\zeta_{c+1} \;=\; \kappa_c\,(D+2-x) \;=\; \kappa_c\,\sqrt{\Lambda}\,\tan(\theta^*/2)$$

where $\tan(\theta^*/2) = \sqrt{\Lambda}/(D+2+x)$. Comparison with the majorant from the tangent addition formula then gives

$$\zeta_{c+\ell} \;\le\; \sqrt{\Lambda}\,\tan\big((\ell - \delta/2)\,\theta^*\big), \qquad \delta \;=\; D \bmod 2,$$

as long as the argument stays below $\pi/2$. More precisely, the path is $j = c, c+1, \dots, s$ when D is even and $j = c+1, c+2, \dots, s$ when D is odd. On this path the induction invariant is

$$p_j \neq 0, \qquad 0 \le \zeta_j \le \sqrt{\Lambda}\tan\big((j - D/2)\theta^*\big) < x.$$

The displayed anchors establish its base case. If it holds at j, then $x - \zeta_j > 0$, so the (p,d) system makes p_{j+1} nonzero with the same sign as p_j; monotonicity of the comparison map gives the next upper bound, while its positive numerator gives $\zeta_{j+1} \ge 0$.

5. *(angular bound)* Since $x > 0$ on the gap (Corollary 3.4), $\theta^* := \arctan(\sqrt{\Lambda}/x)$ lies in $(0, \pi/2)$, and $\sin^2\theta^* = \Lambda/(D+2)^2$. By Lemma 3.6,

$$\begin{aligned}(u+1)^2\sin^2\theta^* \;&<\; \frac{7}{6}\,\frac{D+3}{2Q+3}\,\frac{4(Q+1)(M+1)}{(D+2)^2} \\ &\le\; \frac{7}{3}\,\frac{(M+1)(D+3)}{(D+2)^2} \;<\; \frac{7}{3},\end{aligned}$$

using $(Q+1)/(2Q+3) \le \frac{1}{2}$ and $(D+2)^2-(M+1)(D+3) = Q(D+3)+1 > 0$. We claim that $(u+2)\theta^*/2 < \pi/2$. For $u=1$ this is $\theta^* < \pi/3$, which holds because $\sin^2\theta^* < 7/12 < 3/4$. For $u=2$ it is $\theta^* < \pi/4$, which holds because $\sin^2\theta^* < 7/27 < 1/2$. For $u \ge 3$, Jordan's inequality ($\theta \le \frac{\pi}{2}\sin\theta$ on $[0,\pi/2]$) and $(u+2)/(u+1) \le \frac{5}{4}$ give

$$\frac{u+2}{2}\theta^* \;<\; \frac{u+2}{u+1}\,\frac{\pi}{4}\sqrt{\frac{7}{3}} \;\le\; \frac{5}{4}\,\frac{\pi}{4}\sqrt{\frac{7}{3}} \;<\; \frac{\pi}{2},$$

the last inequality being $25\cdot 7/3 < 64$. Write $j = c+\ell$; then $2j-D = 2\ell-\delta$, and at the target $\ell-\delta/2 = u/2$. Therefore, at every step through $j=s$,

$$(\ell-\delta/2)\theta^* \le \frac{u}{2}\theta^* < \frac{\pi}{2}-\theta^*.$$

Since $x = \sqrt{\Lambda}\cot\theta^* = \sqrt{\Lambda}\tan(\pi/2-\theta^*)$, the comparison bound is strictly below x. This closes the induction invariant at every index through the target; in particular, no denominator in the comparison map vanishes or changes sign. Consequently p retains its sign and $p_{j+1}d_j \le 0$ from the first top-half cell through $j=s$. Steps (1) and (2) now give $T_s \ge 0$. ∎

5. Gap cells with odd Q

Proposition 5.1 *(offset one).* *If Q is odd and $2s = D+1$, then*

$$(D+3)^2(D+5)\,T_s \;=\; 32\,(M+1)(Q+2)\,p_s^2 \;\ge\; 0.$$

Proof. Reflection gives $p_{s-1} = p_{D-s} = -p_s$. Since $D+1-s = s$, the recurrence at index s gives $(s+1)p_{s+1} = (x+s)p_s$, and at index $s+1$ it gives $(s+2)p_{s+2} = xp_{s+1} + (s-D)p_s$. Substituting both into (1.2) and using $2(s+1) = D+3$ and $2(s+2) = D+5$ gives the identity. ∎

For each top-half index j with $p_j \ne 0$, define the scaled ratio

$$y_j := \frac{(D+2)p_{j+1}}{p_j}.$$

At a cell s with $p_s \ne 0$, the quadratic-form identity (3.1) gives

$$G(y_s)p_s^2 = (s+2)(D+1-s)(D+2)^2T_s,$$

where

$$G(y) = u(s+2)y^2 - x(u+1)(D+2)y + (u+2)(D+1-s)(D+2)^2. \tag{5.1}$$

Since $u \ge 1$, this is an upward parabola, and its discriminant is $(D+2)^2q(U)$. On the gap it therefore has two real roots $y_-(u) < y_+(u)$; we call the open interval between them the *gap window*. Thus $T_s \ge 0$ whenever $y_s \ge y_+(u)$. For odd Q, the proof propagates the bound $y_j \ge y_+(2j-D)$ through the recurrence; Lemma 5.2 gives y_+ (the quadratic formula with $s = (D+r)/2$ in (5.1)) and the transition inequality.

Lemma 5.2 *(odd-transition certificate).* *Let $Q \geq 1$, $x = D - 2Q \geq 0$ and $u \geq 1$, and suppose*

$$7(D+3) \;\geq\; 30(u+3)^2.$$

For each offset r for which the following square root is real, put

$$\begin{aligned} g_r \;&=\; \sqrt{x^2(r+1)^2 - r(r+2)\big(E-(r+1)^2\big)}, \\ y_+(r) \;&=\; \frac{(D+2)\big(x(r+1)+g_r\big)}{r(D+r+4)}, \qquad \kappa(r) \;=\; \frac{D-r}{D+r+4}, \\ \Phi_r(y) \;&=\; (1+\kappa(r))x - \kappa(r)\frac{(D+2)^2}{y}. \end{aligned} \tag{5.2}$$

If the windows at u and $u+2$ are both real, then

$$y \geq y_+(u) \Longrightarrow \Phi_u(y) \geq y_+(u+2), \qquad \Phi_u(\infty) \;=\; (1+\kappa(u))x \;\geq\; y_+(u+2).$$

In addition, when D is odd and the window at 1 is real,

$$\frac{(D+2)(2x+D+1)}{D+3} \;\geq\; y_+(1).$$

Proof. Put

$$\begin{aligned} W_1 \;&=\; x(u+1)+g_u, \quad K \;=\; \frac{x\big((u+1)D+u^2+9u+12\big)}{D+u+4}, \\ B_0 \;&=\; u(D-u), \quad C_0 \;=\; (u+2)(D+u+6), \\ \mathrm{disc}_2 \;&=\; x^2(u+3)^2 - (u+2)(u+4)\big(E-(u+3)^2\big), \\ X_{\min 2} \;&=\; \frac{(u+2)(u+4)\big(E-(u+3)^2\big)}{(u+3)^2}, \qquad X_{\max} \;=\; (D-2)^2. \end{aligned}$$

Writing $g_{u+2} = \sqrt{\mathrm{disc}_2}$, direct substitution gives

$$\begin{aligned} \big(\Phi_u(y_+(u)) - y_+(u+2)\big)\frac{W_1 u(D+u+4)(u+2)(D+u+6)}{D+2} \\ \;=\; u(D+u+4)\big(KW_1 - B_0C_0 - g_{u+2}W_1\big). \end{aligned} \tag{5.3}$$

The hypothesis gives $D > u$ and $E > (u+1)^2$. If $x = 0$, then $g_u^2 = -u(u+2)\big(E-(u+1)^2\big) < 0$; hence $x > 0$, so $W_1 > 0$, and every cleared factor is positive. Moreover $y_+(u) > 0$ and

$$\Phi_u'(y) \;=\; \kappa(u)\frac{(D+2)^2}{y^2} \;>\; 0,$$

so it is enough to certify this boundary value.

After reducing g_u^2 to its defining polynomial, clear the denominator of K and define P_1, R_1 uniquely by

$$(D+u+4)^2\big[(KW_1 - B_0C_0)^2 - \mathrm{disc}_2\, W_1^2\big] \;=\; 4(u+2)\big(P_1 + xR_1 g_u\big);$$

the left-hand side is a polynomial because $(D+u+4)K$ is, and the clearing factor is a positive square. As polynomials in $X = x^2$, P_1 is concave, with leading coefficient $2(u+1)^2(u-D)(D+u+6) < 0$, and R_1 is decreasing, with coefficient $2(u+1)(u-D)(D+u+6) < 0$. Consider

$$O_1 = R_1(X_{\max}), \qquad O_2 = P_1(X_{\max}),$$
$$O_3 = (u+3)^4 P_1(X_{\min 2}), \quad O_4 = \frac{(D+u+4)^2}{u+2}(K^2 - \mathrm{disc}_2)\Big|_{X=X_{\max}}.$$

The hypothesis region is parametrized by $D = \frac{30}{7}(u+3)^2 - 3 + v$ and $u = 1 + w$ with real $v, w \geq 0$. After this substitution each O_i is a polynomial in (w, v) with nonnegative rational coefficients:

	O_1	O_2	O_3	O_4
# terms	24	48	77	30
min coefficient	2	10	1	1

Hence $O_1, \ldots, O_4 \geq 0$ on the region. The reality of the successor window and $x \leq D-2$ place $X = x^2$ in $[X_{\min 2}, X_{\max}]$. Thus concavity gives $P_1 \geq 0$ at both ends of the allowed X-interval and hence throughout it, while monotonicity gives $R_1 \geq 0$. Next, $Kx(u+1)$ is increasing in X, because its numerator $X(u+1)\big((u+1)D + u^2 + 9u + 12\big)$ has positive coefficients; the same numerator and $x > 0$ show $K > 0$. The reality of the current window is $X \geq X_b := u(u+2)\big(E - (u+1)^2\big)/(u+1)^2$, and at $X = X_b$ the value of $Kx(u+1) - B_0C_0$, multiplied by the positive factor $(u+1)(D+u+4)/(u(u+2))$, is $4(D+3)(u+2)(D+u+4)$. Hence $Kx(u+1) - B_0C_0 > 0$ for every allowed X, so the squared inequality has the correct unsquared sign and proves the implication for finite y. Finally, after multiplication by the positive factor in O_4, $K^2 - \mathrm{disc}_2$ decreases linearly with X, with coefficient $4(u-D)(D+u+6) < 0$. Thus O_4 and $K > 0$ prove the state at infinity.

For the odd-D base, put $B_o = \big(4x + (D+1)(D+5)\big)/(D+3) > 0$. Direct substitution gives

$$\Big(\frac{(D+2)(2x+D+1)}{D+3} - y_+(1)\Big)\frac{D+5}{D+2} = B_o - g_1,$$
$$(D+3)^2\big(B_o^2 - g_1^2\big) = 4(D+1)(D+5)\big((D+3)^2 - (x-1)^2\big),$$

and the last expression is positive because $0 \leq x \leq D$. ∎

Theorem 5.3 *(gap cells with odd Q). After relabeling so that $Q \leq M$, every gap cell with Q odd and $u \geq 2$ has $T_s \geq 0$.*

Proof. Since $Q \geq 1$, Lemma 3.6 gives $30(u+1)^2 < 7(D+3)$; also $x = D - 2Q \leq D-2$. For a top-half index j, write $r = 2j - D$. By Lemma 3.5, every positive same-parity offset $r \leq u$ has a real gap window, and every transition $r \to r+2 \leq u$ satisfies $30(r+3)^2 \leq 30(u+1)^2 < 7(D+3)$, as Lemma 5.2 requires. To include zeros without division, define the projective state

$$\mathcal{C}_r := \left\{(a,b) : a \neq 0,\ \frac{(D+2)b}{a} \geq y_+(r)\right\} \cup \{(0,b) : b \neq 0\}.$$

The second set is the single state $y = \infty$, and simultaneous zeros are excluded by Lemma 2.1. On finite states the recurrence induces the Möbius map

$$y_{j+1} \;=\; (1+\kappa_j)\,x - \kappa_j\,\frac{(D+2)^2}{y_j}, \qquad \kappa_j \;=\; \frac{D-j}{j+2},$$

increasing in y on $y > 0$; at offset $r = 2j - D$ it is the map Φ_r of Lemma 5.2. By the window interpretation of (5.1), it is enough to prove $(p_j, p_{j+1}) \in \mathcal{C}_{2j-D}$ from the first noncentral index to s.

Reflection for odd Q is $p_{D-j} = -p_j$. For D even: $p_c = 0$ at $c = D/2$, $p_{c+1} \neq 0$ by Lemma 2.1, and $y_{c+1} = 2(D+2)x/(D+4)$. The base inequality $y_{c+1} \geq y_+(2)$ reduces, after eliminating the square root, to $x^2 \leq (D+4)^2$, which always holds. For D odd: $p_c \neq 0$ because $p_{c+1} = -p_c$, hence $y_c = -(D+2)$ at $c = (D-1)/2$, and $y_{c+1} = (D+2)(2x + D + 1)/(D+3)$. The window at 1 is real by Lemma 3.5, and the base inequality $y_{c+1} \geq y_+(1)$ is the final assertion of Lemma 5.2.

At each subsequent offset, Lemma 5.2 maps both the finite and infinite parts of $\mathcal{C}_r$ into $\mathcal{C}_{r+2}$. Induction therefore gives $(p_j, p_{j+1}) \in \mathcal{C}_{2j-D}$ along the induction path. At the target, either $p_s = 0$ and (3.1) gives $T_s = u\,p_{s+1}^2/(D+1-s) \geq 0$, or $y_s \geq y_+(u)$ and hence $G(y_s) \geq 0$. Thus $T_s \geq 0$. ∎

6. Proof of the Main Theorem

Proof of Theorem 1.1 and Corollary 1.2. Fix $M, Q \geq 0$, put $D = M + Q$, and choose $D/2 \leq s \leq D$. By swapping the exponents if necessary, assume $x = M - Q \geq 0$ (Section 1.4), so that $Q \leq M$. Write $u = 2s - D$ and $U = (u+1)^2$.

1. If $s = D$: $T_D = p_D^2 = 1 > 0$.
2. If $u = 0$: Theorem 3.2.
3. If $x^2 U \leq (U-1)(E-U)$: Theorem 3.1.
4. If $U \geq (D+1)^2 - x^2$: Theorem 3.3.
5. Otherwise the cell is a gap cell (Corollary 3.4). If Q is even, Theorem 4.1 applies. If Q is odd, Proposition 5.1 applies when $u = 1$ and Theorem 5.3 when $u \geq 2$.

Every branch ends in $T_s \geq 0$. The corollary follows by telescoping from $\mathrm{Tur}(D) = 1$ and by the reflection symmetry of Tur. ∎

7. Verification

7.1. Exact replays

The programs replay the identities above in exact rational arithmetic with SymPy [MSP+17]. Apart from identities, the only inequalities established by computer are the obligations

$O_1, \ldots, O_4$ of Lemma 5.2, each certified by expanding a polynomial after the stated substitution and checking the signs of its coefficients. No floating-point comparison, asymptotic estimate, or finite enumeration is used in the proof.

The programs also walk the recurrence exactly over finite ranges, as consistency checks of the induction invariants rather than as proof steps: every even-Q gap cell with $D \leq 400$ for Theorem 4.1, and every odd-Q gap cell with $u \geq 2$ and $D \leq 600$ for Theorem 5.3. Table 1 lists the programs; each link opens the committed log in `certificates/`.

Program / log	Checks
`regime-decomposition`	Theorems 3.1–3.3, Corollary 3.4, Lemmas 3.5–3.6, Proposition 5.1, and the real-argument example of Section 1.1.
`even-minimum-gap`	Theorem 4.1: identities, anchors, angular bound, and orbit walks.
`odd-minimum-gap`	Lemma 5.2 and Theorem 5.3: cleared identities, the obligations $O_1, \ldots, O_4$, the base, the region arithmetic, and orbit walks.
`recurrence-and-small-scan`	Cross-check: the recurrence for $M, Q \leq 30$ and $T_s \geq 0$ directly for $M, Q \leq 90$.

Table 1. Verification programs.

7.2. Lean formalization

The directory `lean/` of the repository contains a formalization in Lean 4 with Mathlib of every numbered statement and displayed mathematical claim of this paper. The file `Kraw/Paper.lean` indexes them by number, from Theorem 1.1 and Corollary 1.2 through Proposition 1.3, the statements of Sections 3–5, and the table in Lemma 5.2. There p_s is the coefficient of w^s in $(1+w)^M(1-w)^Q \in \mathbb{Z}[w]$, Schur polynomials are generating functions of semistandard tableaux, and the irreducible polynomial representation of Proposition 1.3 is constructed as the kernel of the map $\Lambda^s V \otimes \Lambda^s V \to \Lambda^{s+1} V \otimes \Lambda^{s-1} V$ that moves one vector from the second factor to the first. Every declaration in the index depends only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`; `make lean` builds the formalization and checks this.

7.3. Code availability

This manuscript is Version 2.0.3. Its source, the verification programs, and their deterministic logs are maintained in the project repository [Pol26], where `certificates/MANIFEST.sha256` identifies all files byte-for-byte. The file `certificates/PAYLOAD.sha256` hashes the verification programs, their logs, the Lean sources, and the dependency lock without the manuscript; its SHA-256 digest is

`83eb74b3065bd4c432ae230b5dda24af7ac454b053ef959f7288de61e2f6173b`

The programs require Python 3 and SymPy 1.12, and the formalization Lean 4 and Mathlib at the versions pinned in `lean/`; the commands are listed in `paper/README.md`. The verification code is available under the BSD 3-Clause License, and the manuscript under the Creative Commons Attribution 4.0 International License.

8. Related work

The classical Turán inequality for Legendre polynomials, discussed by Szegő [Sze48], is $P_n(x)^2 - P_{n-1}(x)P_{n+1}(x) \geq 0$ on the orthogonality interval; its descendants concern the *positivity* of Turán determinants as functions of the argument with the degree fixed, for broad classes of orthogonal polynomials [Sze75].

Recurrence-based criteria. Szwarc [Szw98] gave criteria in terms of monotonicity of the three-term recurrence coefficients. Bustoz and Ismail [BI97, Section 2, Theorem 1, (2.3), p. 2] gave weighted sum-of-squares representations for Turán determinants of symmetric three-term recurrences. Berg and Szwarc [BS09] proved two-sided bounds on normalized Turán determinants, Szwarc [Szw21] extended the positivity criteria, and Krasikov [Kra11] developed Turán inequalities for recurrences with monotone coefficients. Recent criteria and nonnegative representations, including propagation of positivity between determinant indices, are developed in [Kah26]. These results concern the sign of Tur under recurrence and normalization hypotheses, whereas Theorem 1.1 compares adjacent determinants in the ordinary coefficient normalization (Section 1.3).

Krawtchouk bounds. Krasikov (partly with Litsyn) developed Turán-type and Laguerre-type inequalities for discrete orthogonal polynomials and used them for sharp, parameter-uniform bounds on binary Krawtchouk polynomials, their extreme zeros, and Christoffel–Darboux kernels [KL96, Kra01, Kra03, Kra05]. The inequality recorded in [Kra05, Theorem 5, (18), p. 79] gives a fourth-degree refinement coupling two adjacent Turán determinants; Krasikov credits this refinement to Mařík and its extension to the full Laguerre–Pólya class to Dimitrov. That framework is written for exponential coefficients $\sum_k u_k z^k/k!$, so converting it to (1.1) introduces a degree-dependent factorial normalization, and its conclusion concerns a product of adjacent determinants rather than their difference.

Computer algebra proofs. Gerhold and Kauers [GK06, Section 2] prove a Turán inequality by induction and cylindrical algebraic decomposition.

Specialized Schur functions and character values. The factorization theorem of Littlewood and Prasad, presented in [AK22, arXiv version 3, Theorem 2.5], treats general linear characters under balanced root-of-unity specializations, and Karmakar [Kar25, arXiv version 1, Theorem 4.1] gives formulas at arbitrary order-two conjugacy classes; outside the factorized cases these are alternating sums of products of dimensions. Reiner and Stanton [RS98] prove symmetry and unimodality for certain differences of principally specialized Schur functions $s_\lambda(1, q, \ldots, q^n)$, with varying shapes rather than the fixed $\{1, -1\}$ alphabet and rectangle-height sequence considered here.

Other directions. Turán inequalities for coefficient sequences and Jensen polynomials have a separate classical theory [CC89], and uniform asymptotics for Krawtchouk polynomials in the oscillatory regime are available [LW00]. Binary Krawtchouk polynomials at

integer arguments are the spectral kernel of the Hamming scheme [Del73, Lev95, MS77].

To our knowledge, Theorem 1.1 has not previously been stated for the coefficient normalization of $(1+w)^M(1-w)^Q$ at integer (M,Q), equivalently for the ordered rectangular character values in (1.3).

9. Further questions

1. **A tangent comparison for odd Q.** For even Q the central anchor $\zeta_c = 0$ lets the tangent comparison of Theorem 4.1 run without computer assistance. For odd Q the anchor is the state at infinity, and Theorem 5.3 instead propagates the upper window endpoint, using the four transition certificates of Lemma 5.2. Is there a comparison map for odd Q that avoids them?

2. **Beyond the symmetric weight.** For which $a, b > 0$, with $a \neq b$, and which coefficient normalization does an analogous Turán-step monotonicity statement hold for $p_s(a,b) = [w^s]\,(1+aw)^M(1-bw)^Q$ for all integers $M, Q \geq 0$? More generally, which conditions on the locations and multiplicities of positive and negative real roots permit such a statement for other real-rooted products?

3. **Combinatorial meaning of $\mathrm{Tur}(s) \geq 1$.** Is there a lattice-path or coding-theoretic injection realizing $p_s^2 - p_{s-1}p_{s+1} - 1$ as a counting quantity at integer arguments?

4. **Equality away from the center.** Theorem 3.2 classifies central equality. Are there any noncentral cells with $T_s = 0$?

AI-use disclosure

GPT-5.6 Sol assisted with the proof architecture of version 1. Claude Opus 5.5 assisted with the restructured proof and verification code of version 2, and GPT-6 Luna agents wrote most of the Lean formalization under its direction. The author selected and checked the final arguments and is responsible for all mathematical claims, citations, computations, and wording.

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